

Quiz 11

Instructions: You must show all of your work to receive credits for correct answers.

Problem 1. (6 points) Find and graph the corresponding Cartesian equation for each of the following parametric equations:

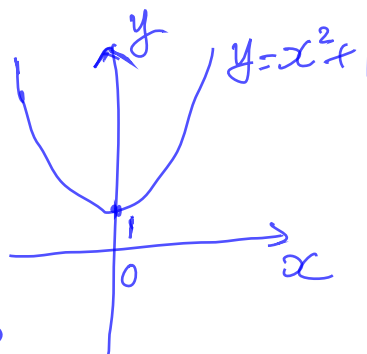
(a) $x = 3t$, $y = 9t^2 + 1$, $-\infty < t < \infty$.

$$x = 3t \Rightarrow t = \frac{x}{3}$$

$$\text{Thus } y = 9t^2 + 1$$

$$y = 9 \cdot \left(\frac{x}{3}\right)^2 + 1$$

$$\Rightarrow \underline{y = x^2 + 1}, \quad -\infty < x < \infty$$



(b) $x = 2 \cos t + 1$, $y = 2 \sin t - 2$, $0 \leq t \leq 2\pi$.

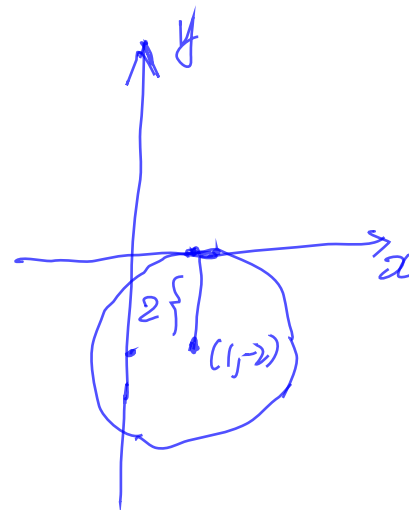
$$x = 2 \cos t + 1 \Rightarrow x - 1 = 2 \cos t$$

$$y = 2 \sin t - 2 \Rightarrow y + 2 = 2 \sin t$$

$$\Rightarrow (x-1)^2 + (y+2)^2 = (2 \cos t)^2 + (2 \sin t)^2$$

$$= 4(\cos^2 t + \sin^2 t)$$

$$\Rightarrow \underline{(x-1)^2 + (y+2)^2 = 4}$$



Problem 2. (4 points) Find an equation for the line tangent to the curve $x = 2 \sin t$, $y = 4 \cos t$ at the point defined by $t = \pi/4$.

$$\frac{dx}{dt} = \frac{d}{dt}(2 \sin t) = 2 \cos t, \quad \frac{dy}{dx} = \frac{d}{dt}(4 \cos t) = -4 \sin t$$

$$\Rightarrow \frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{-4 \sin t}{2 \cos t} = -2 \tan t$$

$$\Rightarrow m = \left. \frac{dy}{dx} \right|_{t=\pi/4} = -2 \tan\left(\frac{\pi}{4}\right) = -2$$

$$\text{The point defined by } t = \pi/4 \text{ is } \left(2 \sin\left(\frac{\pi}{4}\right), 4 \cos\left(\frac{\pi}{4}\right)\right) = (\sqrt{2}, 2\sqrt{2})$$

$\Rightarrow$ The equation for the tangent line is

$$y - 2\sqrt{2} = -2(x - \sqrt{2}) \Rightarrow \underline{y = -2x + 4\sqrt{2}}$$